ESPaperPlay / DEVELOPMENT BOARD r4

ESP32-S3-WROOM-1-N16R8 / GDEY075T7-T01 / protected 1-cell battery

Nine functional sheets. Schematic review stage; no PCB or manufacturing release.

POWER_INPUT_CHARGER

Input / charger / battery

文件: POWER_INPUT_CHARGER.kicad_sch

POWER_3V3_CONTROL

Regulator / IO expander / gauge

文件: POWER_3V3_CONTROL.kicad_sch

MCU_CORE

N16R8 / reset / user keys

文件: MCU_CORE.kicad_sch

USB_INPUT

USB-C / protection / source select

文件: USB_INPUT.kicad_sch

POWER_DOMAINS

Five switched power domains

文件: POWER_DOMAINS.kicad_sch

TOUCH_ISOLATION

GT911 four-signal isolation

文件: TOUCH_ISOLATION.kicad_sch

EPD_DISPLAY

UC8179 / boost / 24-pin FPC

文件: EPD_DISPLAY.kicad_sch

STORAGE_SD

microSD / bus isolation / detect

文件: STORAGE_SD.kicad_sch

AUDIO

I2S amplifier / microphone

文件: AUDIO.kicad_sch

ASSEMBLY / REVIEW BOUNDARIES

R108 CHG_ARM stays DNP. Battery / NTC / charging window are not approved.
J502 source select and J202 MAIN_RUN each use ONE shunt only.
SD_IO_EN uses IOX P13: power valid -> connect I/O; unmount -> isolate -> power OFF.
Candidates assigned; cable mating, biased capacitance, thermal tests and procurement remain open.
No measured power / thermal / USB / ESD performance; see the project README before assembly.

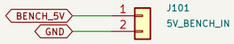
hardware/devboard-r0/README.md / docs/v2/hardware/power-design-spec.md

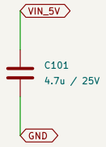
DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
ESPaperPlay / xiaobaoaaaaaa

Sheet: /		
File: ESPaperPlay_Devboard.kicad_sch		
Title: ESPaperPlay DEVELOPMENT BOARD — PERIPHERAL DRAFT		
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 1/10

1 POWER PATH AND PROTECTED BATTERY INTERFACE

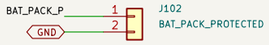
5 V only. USB_INPUT selects protected USB or J101 BENCH_5V into VIN_5V.

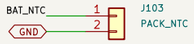




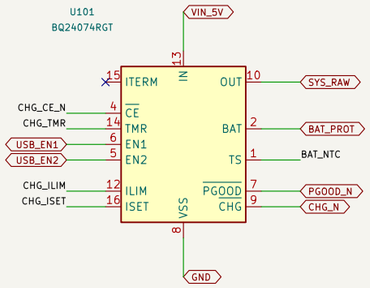
Known, current-limited 5 V bench source.
Select at J502; never bridge both sources.

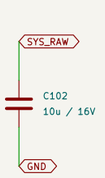


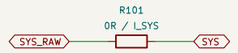




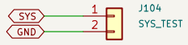
755570 / 4000mAh / 10k NTC advertised.
3.5mm bench terminals; harness/NTC unqualified.
No fixed resistor in place of pack NTC.







R101: measure total SYS load.
CE pull-up stays on SYS_RAW side.











4000mAh: 297mA / 9h typical is NOT a qualified charge profile.
R108 stays DNP. ITERM15 NC; pack/NTC data required.

3 AON-SAFE VBUS SENSE



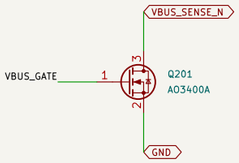


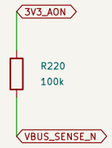




PWR_FLAG: normal RUN with R101/R203 populated.
ERC does not model removable links or input protection.



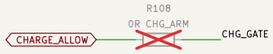


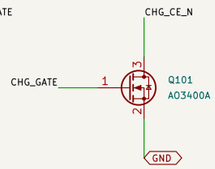


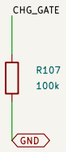


LOW = protected USB branch powered.
PGOOD reports selected input; PTC trip differs.

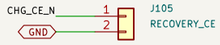
2 HARDWARE CHARGE INHIBIT / CURRENT MODE / STATUS







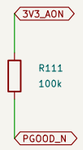


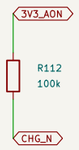






EN2 EN1: 00=100mA, 01=500mA,
10=RILIM, 11=USB suspend.
Do not infer source capacity from VBUS.





J105 normally OPEN. Recovery only: MAIN_RUN OFF, AMP disabled,
EN1/EN2=00, qualified pack + NTC, known bench source.
R108 is DNP: firmware cannot enable charging in initial assembly.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

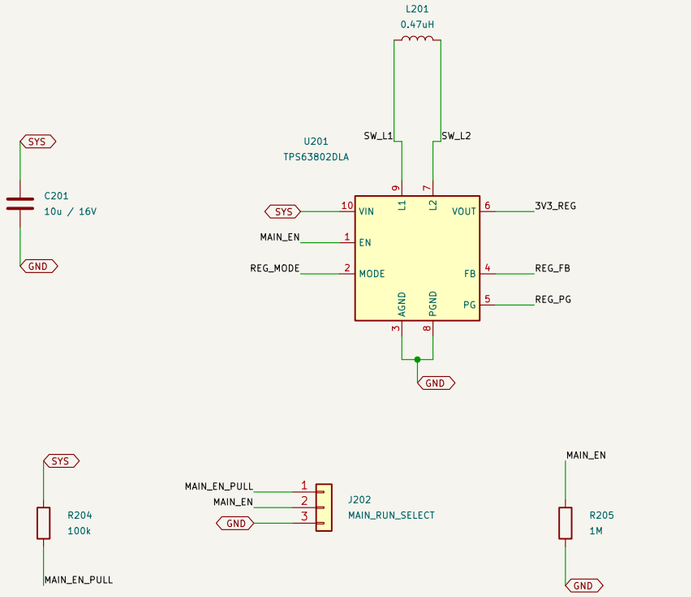
ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_INPUT_CHARGER/
File: POWER_INPUT_CHARGER.kicad_sch

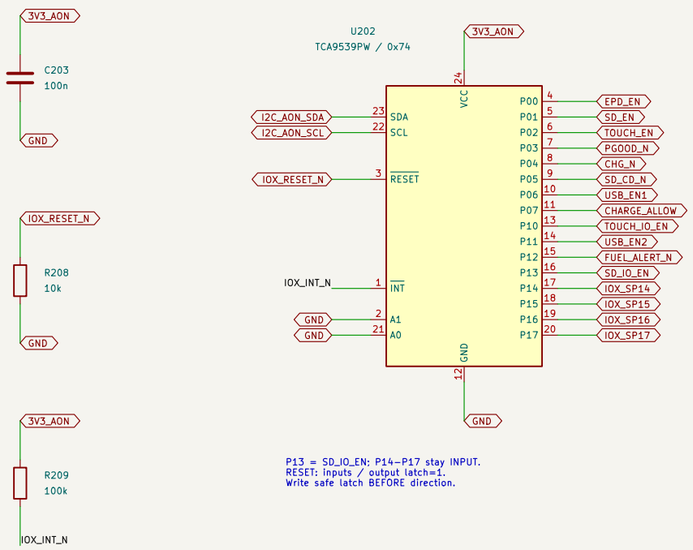
Title: POWER INPUT / BQ24074 CHARGER

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 2/10

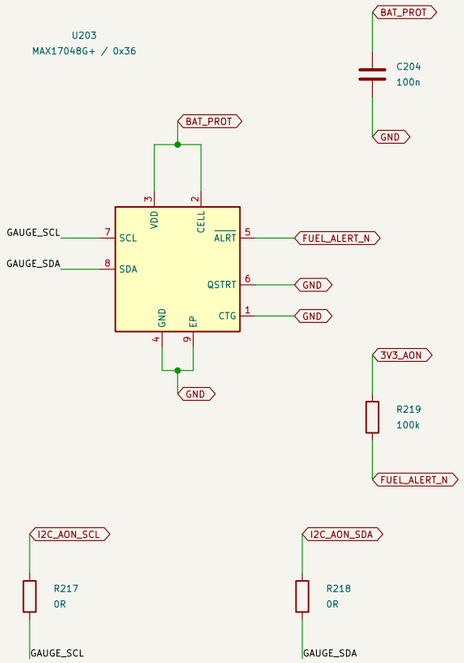
1 MAIN BUCK-BOOST / MEASUREMENT / RECOVERY



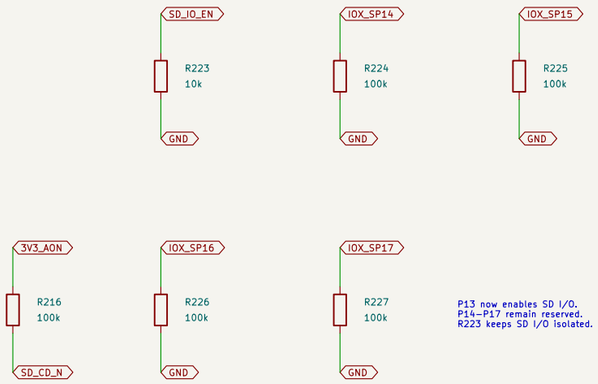
2 AON I2C CONTROL / RESET DEFAULTS



3 BATTERY GAUGE



4 SPARE PORT DEFAULTS



DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

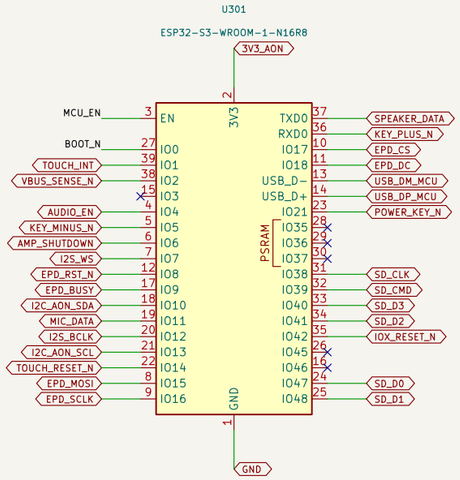
ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_3V3_CONTROL/
File: POWER_3V3_CONTROL.kicad_sch

Title: 3.3 V REGULATOR / POWER CONTROL / GAUGE

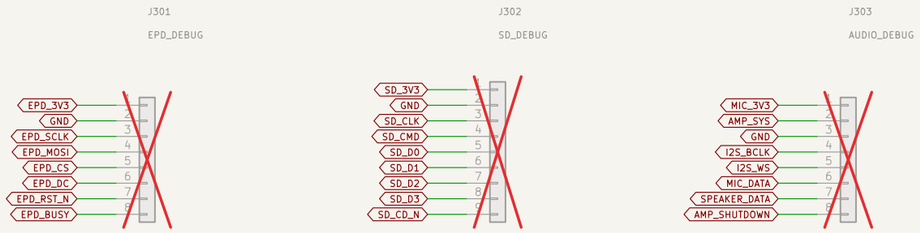
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 3/10

1 MCU / N16R8 PHYSICAL PAD MAPPING



NC: GPIO3 / 45 / 46 (straps); GPIO35 / 36 / 37 (PSRAM).
Do not substitute WROOM-1-N16R16V: pad voltage differs.
Pads 24 / 25 = GPIO47 / 48, respectively.
Antenna at PCB edge; final keepout per module land pattern.

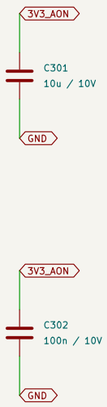
2 MCU-SIDE DEBUG HEADERS / ON-BOARD LOADS CONNECTED

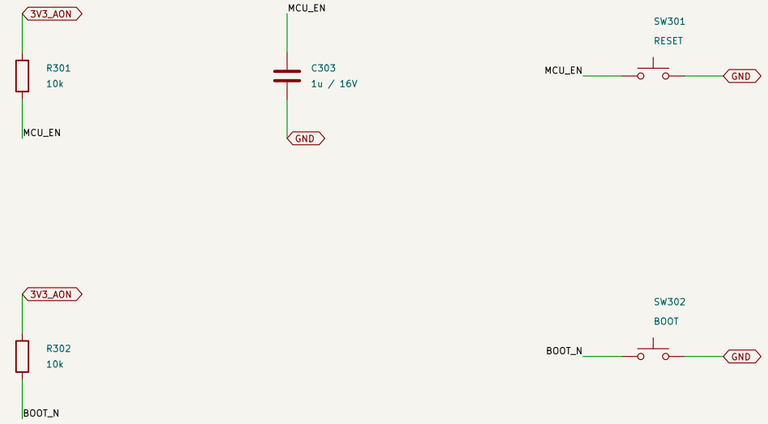


DNP DEBUG: high-impedance probes only; never attach competing peripheral modules.
Audio: MIC_3V3 and AMP_SYS are different supplies; check module voltage.
MCU-side signals: on-board buffers / switches are on each peripheral sheet.
Module pads 1 / 40 / 41 (EPAD) to GND: PCB antenna keepout required.

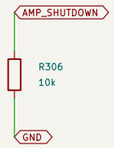
3 MAINTENANCE RESET / DOWNLOAD

LOCAL BYPASS

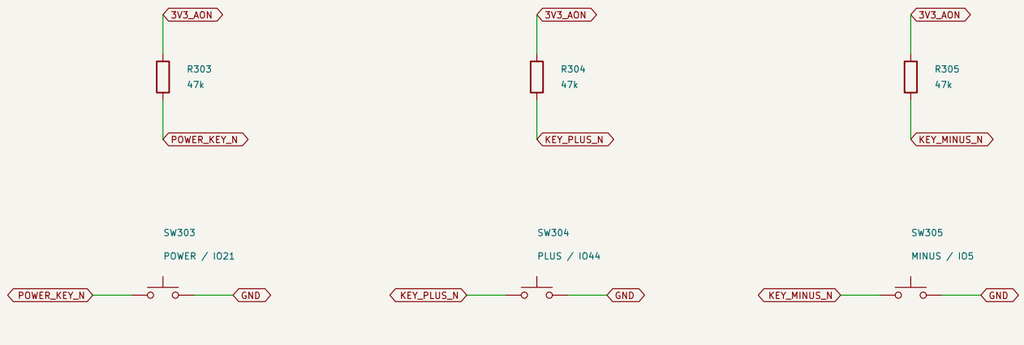




EN: 10k / 1uF initial RC; validate actual power ramp.
BOOT: pull-up only, no large capacitor.
GPIO43 / 44 are assigned: no always-connected UART0.
Native USB Serial/JTAG is the default recovery/debug path.



4 USER KEYS / SOFTWARE DEBOUNCE



IO21 is the normal sleep/wake key. EN is a separate service control.
I2C pull-ups and IOX_RESET_N pull-down live on POWER_3V3_CONTROL.
EPD / SD / touch / audio signals are global labels to their owning sheets.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /MCU_CORE/
File: MCU_CORE.kicad_sch

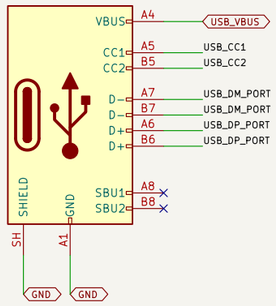
Title: ESP32-S3 N16R8 / RESET / USER KEYS

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 4/10

1 USB-C DEVICE / NO PD / NO HOST POWER

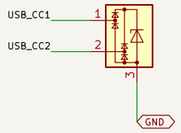
J501

USB-C USB2 16P



U502

TI TPD2E2U06DCKR



USB_CC1

R501

5.1k 1%

GND

USB_CC2

R502

5.1k 1%

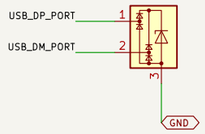
GND

CC1 / CC2 each have their own Rd.
A6+B6 = D+; A7+B7 = D-.
Shield bonded to local GND at connector; final EMC review required.

2 DATA ESD / MCU-SIDE MATCHING

U501

TI TPD2E2U06DCKR



R503

22R





R504

22R



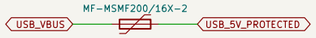


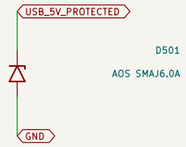
USB_DM_MCU -> GPIO19 / module pad13.
USB_DP_MCU -> GPIO20 / module pad14.
Target 90 ohm differential; short pair, continuous GND return.
USB default 100mA; firmware authorization precedes any 500mA mode.
CC resistors do not negotiate PD, or authorize 1.5A / 3A.
No ESD / USB compliance claim until board-level validation.

3 USB BRANCH PROTECTION / SINGLE-SHUNT SOURCE SELECT

F501

MF-MSMF200/16X-2

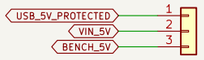






J502

ONE SHUNT: USB / BENCH



1-2 USB / 2-3 BENCH / OPEN = input disconnected.
VIN_5V goes to charger; protected USB is sensed before J502.
One shunt. Known regulated 5V, CURRENT-LIMITED bench input.

PTC / TVS protect the USB branch only: no sustained OVP or automatic source ORing.
TVS leakage / overshoot, fault current, VBUS inrush and cable drop require tests.
F501200/16X: 60C hold1.48A vs1.085A budget; fault/TVS and USB assembly tests remain.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /USB_INPUT/
File: USB_INPUT.kicad_sch

Title: USB-C DEVICE / ESD / 5V SOURCE SELECTION

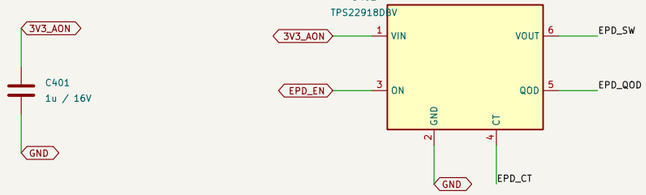
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 5/10

FIVE TPS22918 DOMAINS / COMPONENT VALUES ARE UNVALIDATED CANDIDATES

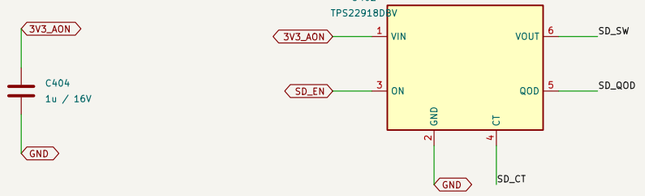
CT = 4.7nF / 50V COG 5%; RQOD = 330R + internal RPD. Links measure downstream load; switch IQ is upstream.

EPD/SD/TOUCH ON pulls: R212/R213/R214 on POWER_3V3_CONTROL. AUDIO_EN: R416 below. No reverse-current blocking.

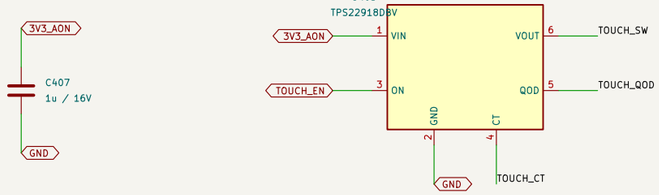
1 EPD



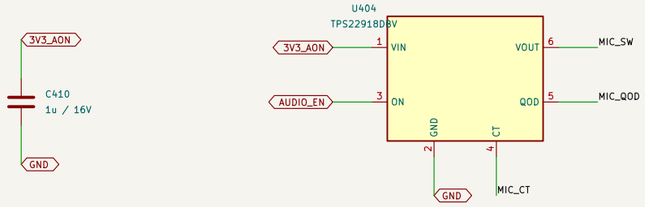
2 SD



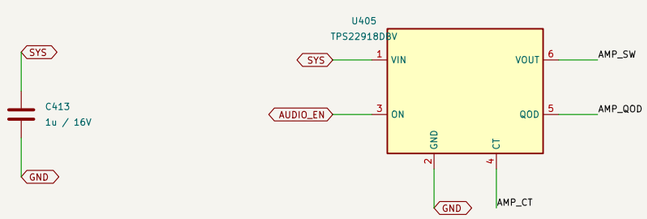
3 TOUCH



4 MIC



5 AMP





R416: both audio rails default OFF when GPIO4 is high impedance.
SYS may remain alive with AON=0; measure AMP leakage in this state.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_DOMAINS/
File: POWER_DOMAINS.kicad_sch

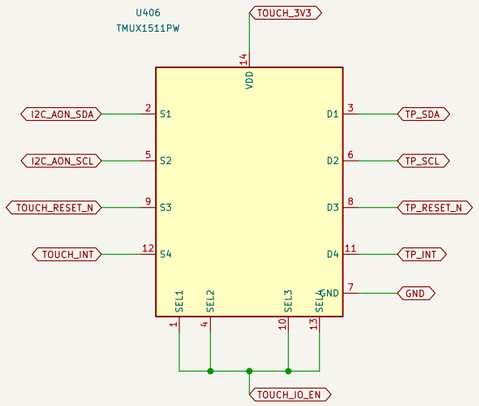
Title: PERIPHERAL POWER / FIVE SWITCHED DOMAINS

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 6/10

1 TMUX1511PW: ALL FOUR SIGNALS ISOLATED / ACTIVE-HIGH SELECT

VDD, bypass, pull-ups and GT911 VCC all use TOUCH_3V3 AFTER R408 (I_TOUCH). Never power the switch from AON.

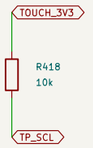




MCU / AON side
GPIO10 SDA; GPIO13 SCL
GPIO14 RESET; GPIO1 INT
R210/R211: 4.7k to AON

R215 = 10k to GND on control sheet.
All SELs LOW: isolated; HIGH: connected.
INT remains bidirectional for GT911 boot.

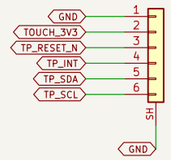












0.5mm FPC6 + GND retention pads.
Keep extension; candidate footprint set.
Check contact face / full-chain pin1.



PWR_FLAG: normal assembly with R408 fitted.
ERC does not model an open measurement link.

2 POWER / SIGNAL SEQUENCING AND MEASUREMENT

ON: keep TOUCH_IO_EN=0; power TOUCH; wait for valid rail; set MCU RESET/INT; connect TMUX; execute full GT911 reset.
OFF: stop access and touch wake; set TOUCH_IO_EN=0; then TOUCH_EN=0. Do not reset IOX merely on one I2C error.
Start I2C at 100kHz. Pull-up estimate is 3.20k with the present 4.7k and 10k; include actual panel resistors and capacitance.
Powered-off protection: signal pins <=3.6V for VDD<1.5V; verify supply ramps and signal leakage with actual hardware.
With AON=0: R215 must pull SEL LOW even while TOUCH retains charge. GPIO hold/reset behavior is not a power guarantee.

3 LOAD-SWITCH REVIEW / MEASUREMENT LIMITS

Target for each low-voltage load rail: <0.2V within 1s after OFF. EPD high-voltage nodes are outside this target.
CT=4.7nF: TI table ~8ms at 3.3V: formula ~8.6ms (3.3V), ~11.5ms (4.4V). Actual load turn-on and inrush need measurement.
QOD=330R external plus internal RPD. Example only: 200uF, 355R, 3.3V -> 0.2V in ~0.20s with VIN still valid.
Input loss weakens QOD; timing is not guaranteed at VIN=0. Do not backfeed a rail while QOD is discharging it.
Remove measurement OR only AFTER stopping the peripheral and isolating its signals. QOD remains connected on the load side.
TOUCH: opening R408 also removes TMUX VDD; I2C pulls remain on that same isolated load rail. Verify residual voltage first.
External rail injection requires upstream OR AND QOD resistor removal, plus reviewed signal isolation; no direct supply paralleling.

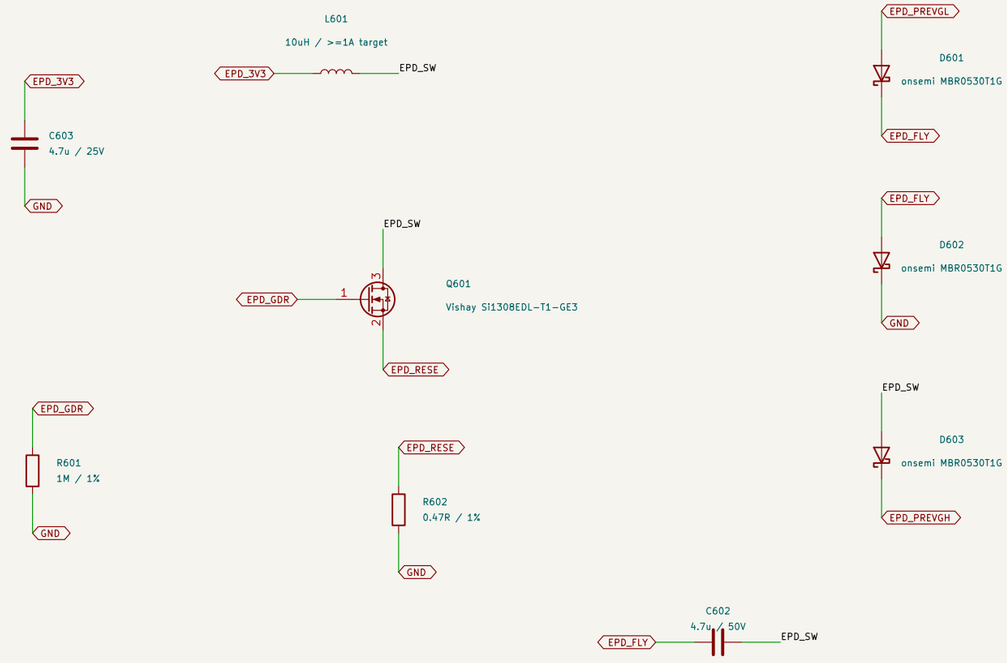
Add actual EPD-driver / SD / audio capacitance before final CT and QOD selection.
SYS -> AMP is a supply gate, not a 5V boost. No load switch provides fault-current limiting.
Refs: TI SLVSD76C pp3,13-15; SCDS390B pp3-4,20-22. Values are NOT bench-validated.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
ESPaperPlay / xiaobaoaaaaaa

Sheet: /TOUCH_ISOLATION/		
File: TOUCH_ISOLATION.kicad_sch		
Title: GT911 FOUR-SIGNAL ISOLATION / DOMAIN TEST NOTES		
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 7/10

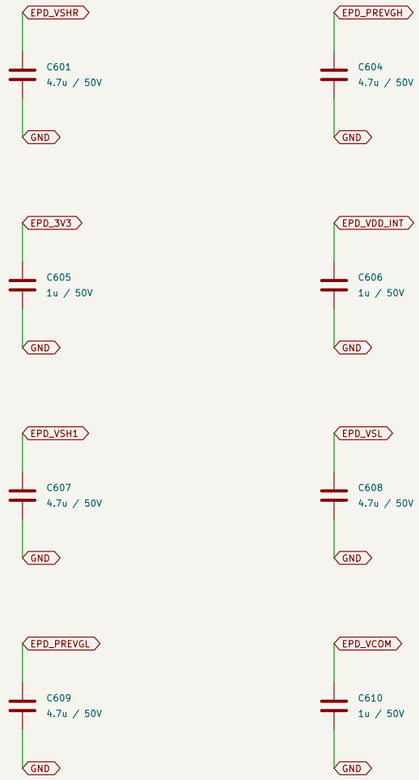
1 GOOD DISPLAY p38 BOOST TOPOLOGY / CANDIDATE PARTS

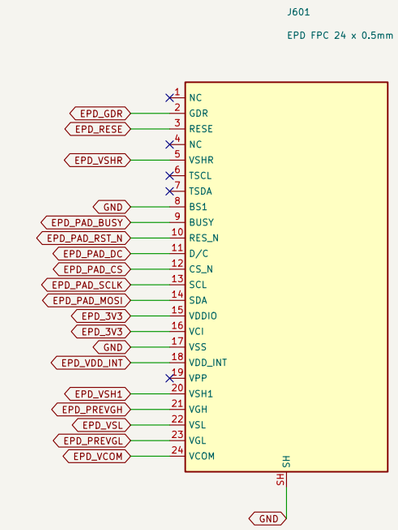
3.3V input only. Panel-generated HV: no external supply injection. Values are reference starting points, not measured.



D601: A=VGL, K=FLY; D602: A=FLY, K=GND; D603: A=SW, K=VGH.
Q601 source is RESE, not direct GND. Keep SW/FLY loops compact.
HV MLCC: X5R/X7R, effective capacitance at bias must be qualified.

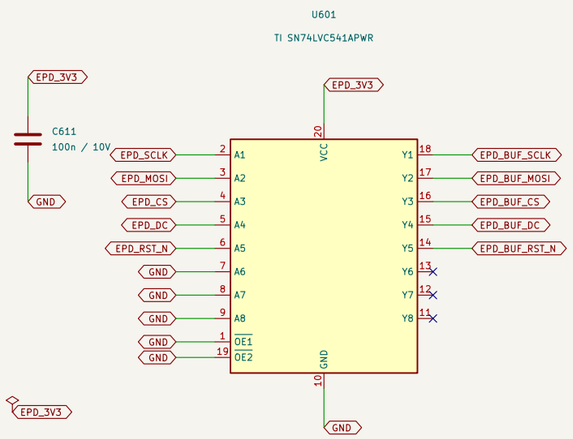
2 PANEL PINS / STORAGE AND DECOUPLING

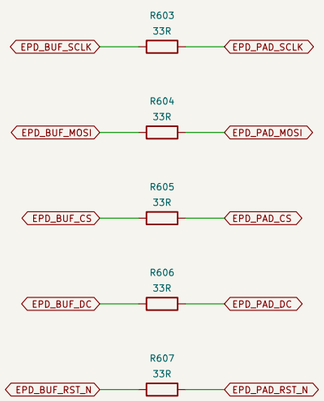




BS1=0: 4-wire SPI, write-only SDI in this board.
NC 1/4/19: unused external-temp pins6/7 left open.
VDD18 is INTERNAL: C606 only: never short to3.3V.
24 signals + SH to GND; check retained extension chain.

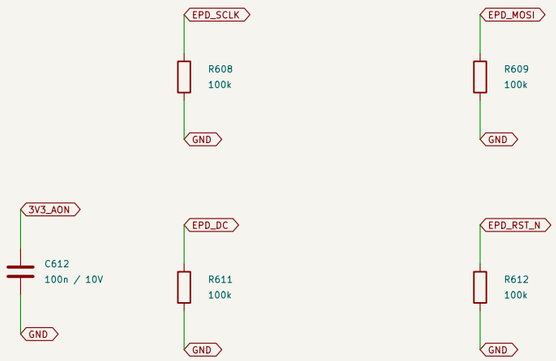
3 FIVE MCU OUTPUTS / EPD-SUPPLIED Ioff BUFFER



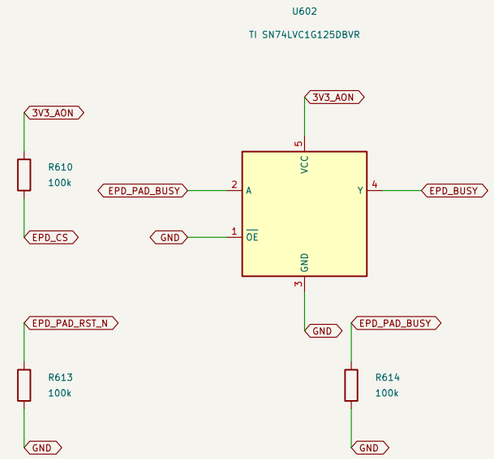


Flag: R402 populated, normal RUN only.
Unused buffer inputs tied low; outputs NC.

4 DEFAULTS / BUSY RETURN



Before ON: CS=1, RESET=0, CLK/MOSI/DC=0; wait valid rail, then reset panel.
Before OFF: finish refresh, sleep panel, hold safe signals; disable EPD supply.
Buffers do not replace reset timing or leakage/ramp testing. BUSY low while OFF is invalid.
No SDI readback. Check +/-HV, gate/current-sense and FPC before connecting the panel.



DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

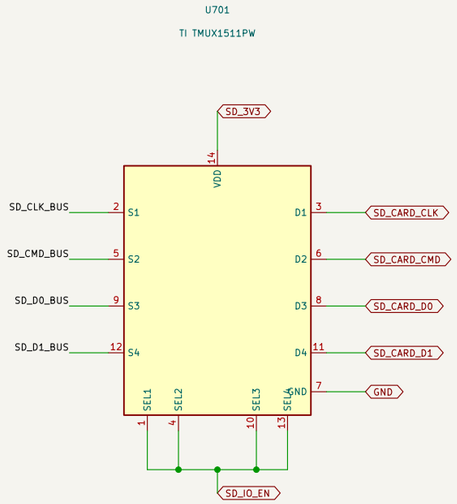
Sheet: /EPD_DISPLAY/
File: EPD_DISPLAY.kicad_sch

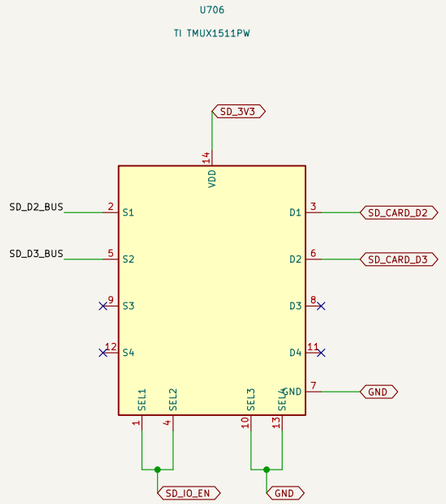
Title: EPD REFERENCE BOOST / 24-PIN PANEL / DOMAIN BUFFERS

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 8/10

1 SIX-LINE BUS ISOLATION / SEPARATE SD_IO_EN CONTROL

Two TMUX1511PW on SD_3V3. IOX P13 controls isolation; P01 controls card power. Six bidirectional paths; no MCU-to-card bypass.







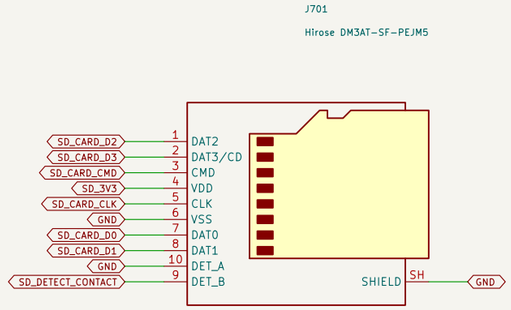












Physical card: 1 DAT2 / 2 DAT3 / 3 CMD / 4 VDD
5 CLK / 6 GND / 7 DAT0 / 8 DAT1.
No 1.8V UHS mode. Begin at 400kHz init, then20MHz target.
CD closes to GND on insertion; debounce in software.

2 CARD-SIDE PULL-UPS / DECOUPLING / DEFAULT DISCONNECT



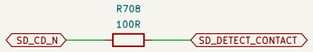


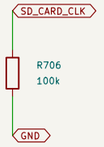






SD_IO_EN (IOX P13): LOW isolates all six lines.
R223=10k default pull-down on control page.
Connect only after SD power has settled.





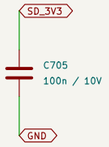






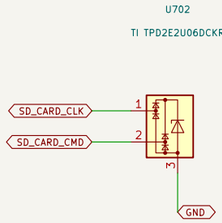


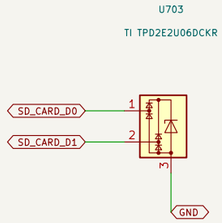
PWR_FLAG assumes R405 fitted.
SD_3V3 is downstream of I_SD.

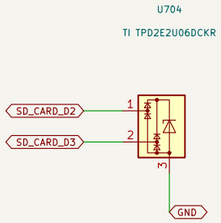


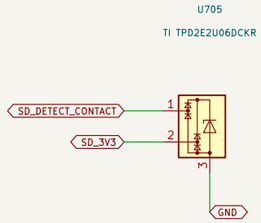


3 CONNECTOR-SIDE ESD / NO SUPPLY CLAMP RAIL









ON: SD_IO_EN=0; enable SD_EN; wait validated rail-settle time; then SD_IO_EN=1 and initialize the card.
OFF: block requests, flush/close/unmount, stop SDMMC; SD_IO_EN=0; then SD_EN=0. Never cut power on unmount timeout.
TMUX: VDD<1.5V allows signal0..3.6V; ramps/functional behavior still need validation. Spare SELs LOW; spare S/D NC.
SD power rail / GPIO hold / removal behavior / 4-bit20MHz signal integrity remain bench tests; no measured throughput claim.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

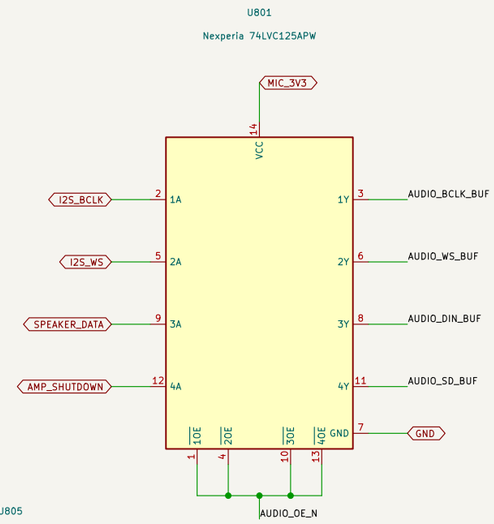
ESPaperPlay / xiaobaoaaaaaa

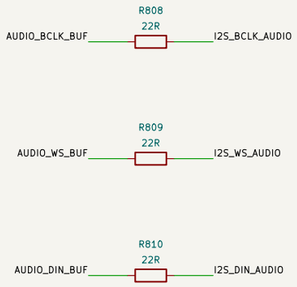
Sheet: /STORAGE_SD/
File: STORAGE_SD.kicad_sch

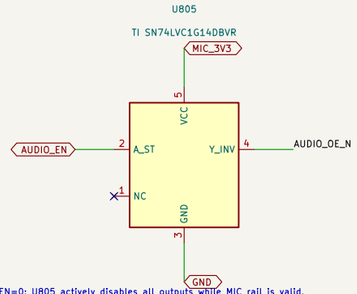
Title: MICROSD / 4-BIT BUS / INDEPENDENT SIGNAL ISOLATION

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 9/10

1 FOUR MCU OUTPUTS / AUDIO_EN HARDWARE DISABLE







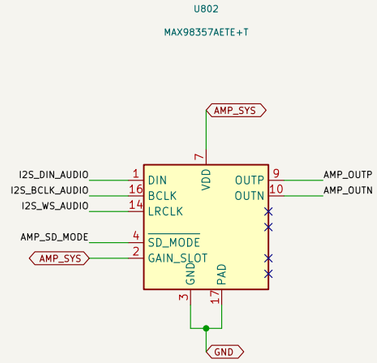


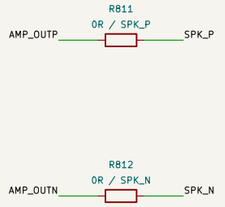


AUDIO_EN=0: U805 actively disables all outputs while MIC rail is valid.
At VCC=0, use the selected vendor Ioff specification; not zero leakage.
R306 already pulls MCU AMP_SHUTDOWN low; R416 pulls AUDIO_EN low.

2 MAX98357A / SYS SUPPLY / LEFT CHANNEL / 6dB







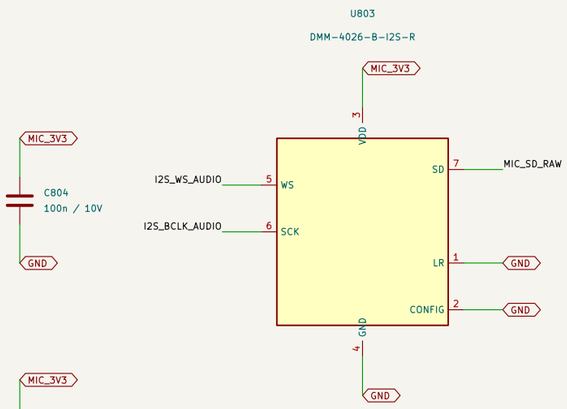


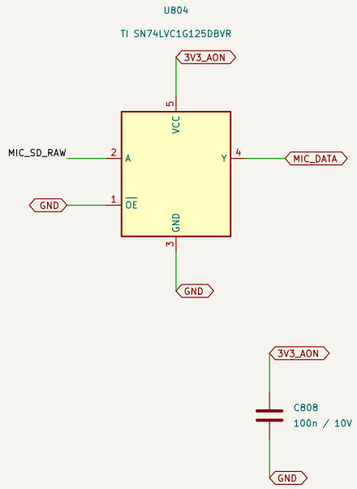




Selected speaker: 3525 cavity / 8 ohm / 2W. Both BTL terminals float from GND.
Initial output cap 0.5W; no 5V boost. Verify low-battery THD/pop and harness.

3 PUI I2S MIC CANDIDATE / DATA RETURN ISOLATION





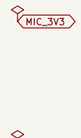
PUI pin map: 1 LR, 2 CONFIG, 3 VDD, 4 GND, 5 WS, 6 SCK, 7 SD.
Both control straps LOW. Footprint is NOT the old six-pad ICS-43434.
AON return buffer remains powered during deep sleep; budget its ICC/leakage.

4 FIRST BRING-UP CONTRACT / NOT MEASURED

Start at 48kHz, two 32-bit I2S slots, BCLK=64fs=3.072MHz.
PUI: 24-bit data / 18-bit precision; LEFT slot. 20ms is a datasheet startup reference only; discard initial frames with measured settling margin.
32/44.1/48kHz are the candidate shared set; 32k is at the MIC clock floor.
The old 16kHz firmware profile is NOT valid for this microphone candidate.
Software mixes/resamples playback into LEFT; no automatic stereo average.

ON: hold AMP_SHUTDOWN low and clocks/data low; enable AUDIO_EN;
wait for both rails; establish clocks/zeros; discard settling frames; unmute.
OFF: mute first, finish audio, stop BCLK without leaving BCLK running
with LRCLK absent; drive signals low, then AUDIO_EN=0.
MIC/AMP discharge rates differ: measure SD_MODE injection and pops.
Mute/stop before opening R411/R414. Never inject external rail power
without reviewing QOD, buffer paths and supply disconnection.

U801/U805 use the MIC rail; U804 remains on AON. Each adds leakage.
Dynamic clocks and pull-down currents belong in the active audio budget.
Actual speaker, MIC lifecycle, acoustic port and EMI remain review gates.



Flags: R411/R414 fitted.
Not a power-fault model.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /AUDIO/
File: AUDIO.kicad_sch

Title: AUDIO / I2S INPUT ISOLATION / MONO PLAYBACK + MIC

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 10/10